

Agentic Coding IDE Extensions Introduction



SoftUni Team
Technical Trainers



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Software University

<https://softuni.org>

Have Questions?

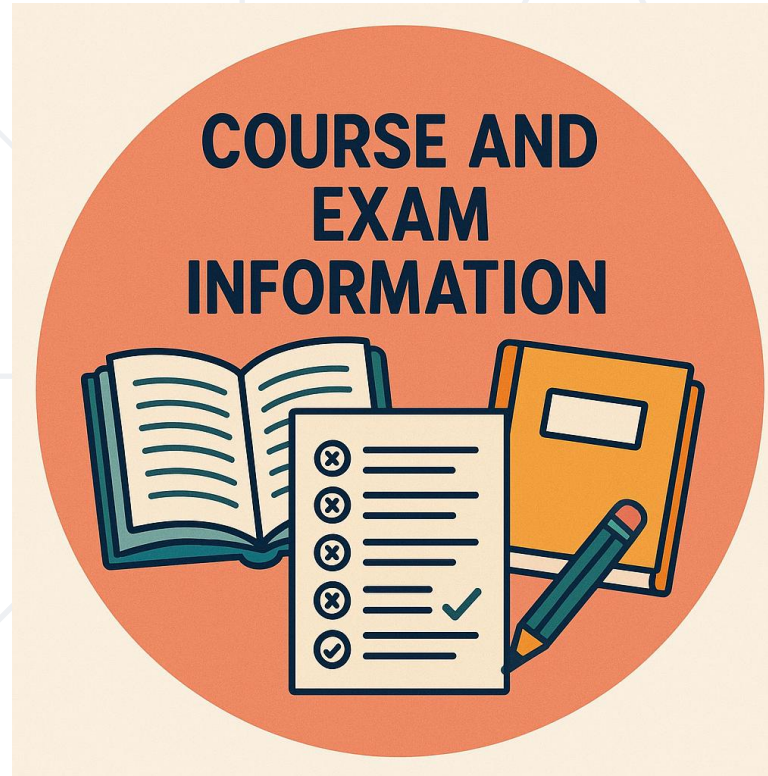
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#AI-Dev

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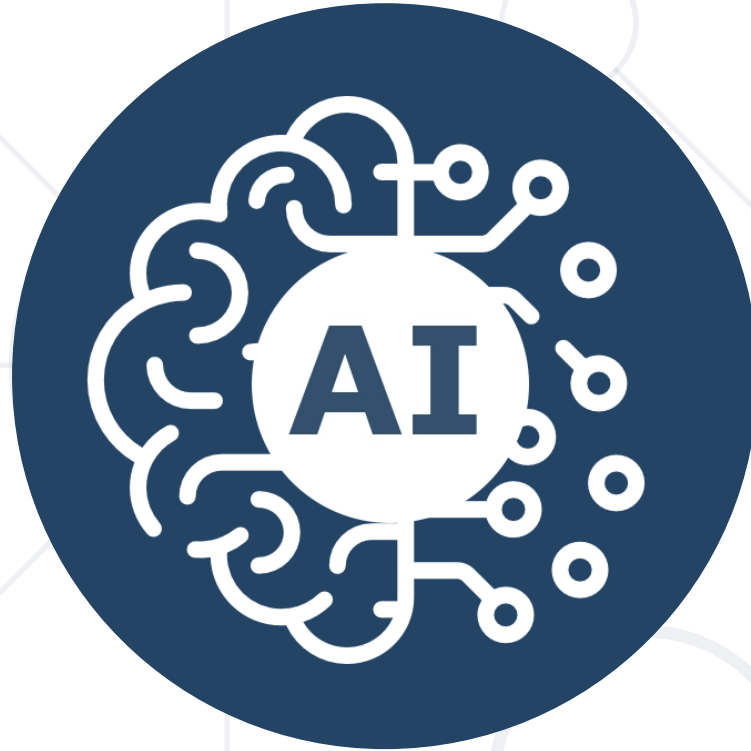
Course and Exam Information

1. Not focused on any specific language
2. Demos will use multiple languages
3. Practices work across all tech stacks



- Create a project on your own topic
- Start now
- Submit until the end of the course

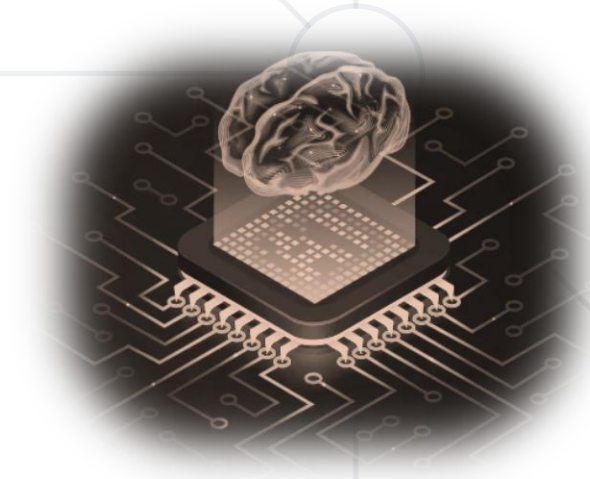




Generative AI and LLMs

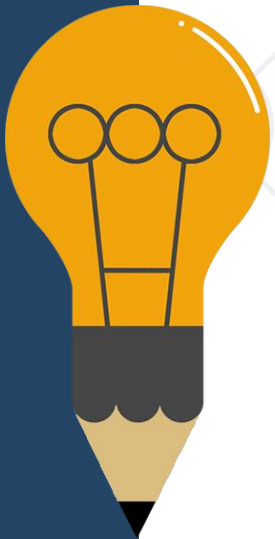
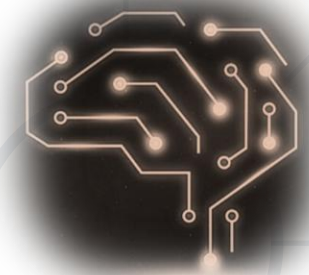
LLMs, Coding Models, Agents, Tokens

- **Artificial intelligence** (AI) == intelligent computer system
 - Performs tasks that typically require **human intelligence**
- **Machine learning**
 - AI uses **algorithms** and mathematical **models** that are trained on huge **data sets**, through many, many **examples**
- Examples of **machine models** (AI models):
 - **GPT-5** generates text (by text prompt)
 - **FLUX 1.1** generates images (by text prompt)
 - **SpeechT5** generates human speech (by text)



Generative Artificial Intelligence

- **Generative Artificial Intelligence** (GenAI)
 - **Large machine models**, trained on human text, images, documents, audio, video
 - **Generate** text, code, images, videos by **prompt**, answer questions, solve problems, create software
 - **Modalities** (what models take as input and output): text, images, screenshots, audio, video, code
 - Require **huge computing resources** for training and a significant resources for inference (generating answers)



- **Large Language Models (LLMs)**
 - **Text content generators**, trained on huge data sets (books, articles, websites, programming code, synthetic data, etc.)
 - Example: **Llama 4 Maverick** is trained on **22T tokens**
 - Learn dozens of **languages**, grammar, text writing and **coding**
 - Fine-tuned to **answer questions** and **follow instructions**
 - Have **parameters** (capacity to remember information and relationships), e. g. **32B**, **405B** or **1.4T** parameters
 - Examples: ChatGPT, Claude, Gemini, DeepSeek, Llama, Mistral

- **Tools** are external commands / services used by LLMs to perform tasks beyond text generation
 - For example, using tools, LLM can run **terminal commands**
 - ChatGPT uses tools to search Internet and draw charts
- **Reasoning models** (thinking models) simulate human-like logical thinking, planning and problem-solving
- **AI dev agents** use tools + reasoning to perform multi-step tasks
 - First, the agent **creates a plan**, using a reasoning model
 - Then, it **executes the plan** step-by-step: writes code, uses tools, runs the code, checks for errors, fixes bugs, etc.



Intro to AI-Assisted Development

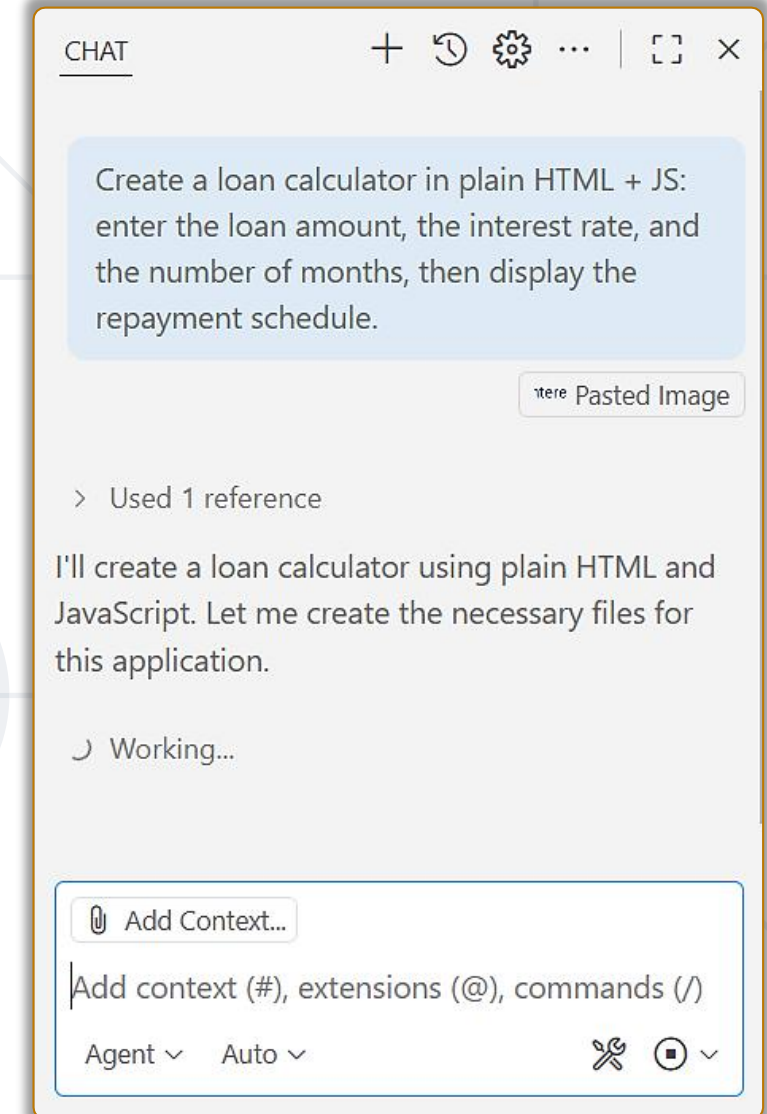
App Development with AI Prompts

What is AI-Assisted Development?

- Create apps using **AI prompts**
 - You write a sequence of **AI prompts** (like in ChatGPT)
 - **AI writes the code** for you, runs, tests, fixes bugs, etc.
- AI-assisted dev environments
 - For **hobbyists**: **Bolt, Lovable, Replit, v0, Firebase Studio**
 - Quite simplified, cloud-based, running in the browser
 - For **professionals**: **GitHub Copilot, Cursor, Claude Code**
 - Installed locally, more complex, quite more powerful



- **GitHub Copilot (+ VS Code)**
 - Professional **AI dev environment** with powerful **AI dev agent**
 - Natively **integrated into VS Code** (the world's most widely used IDE)
 - **Huge popularity**: 68% of developers use Copilot, 20M users ([Sept 2025](#))
 - Chat modes: **Agent** / **Ask** / **Plan**
 - <https://github.com/features/copilot>



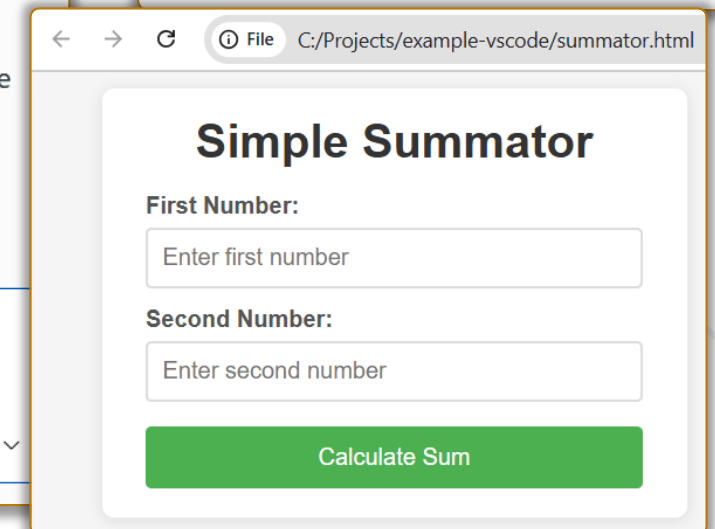
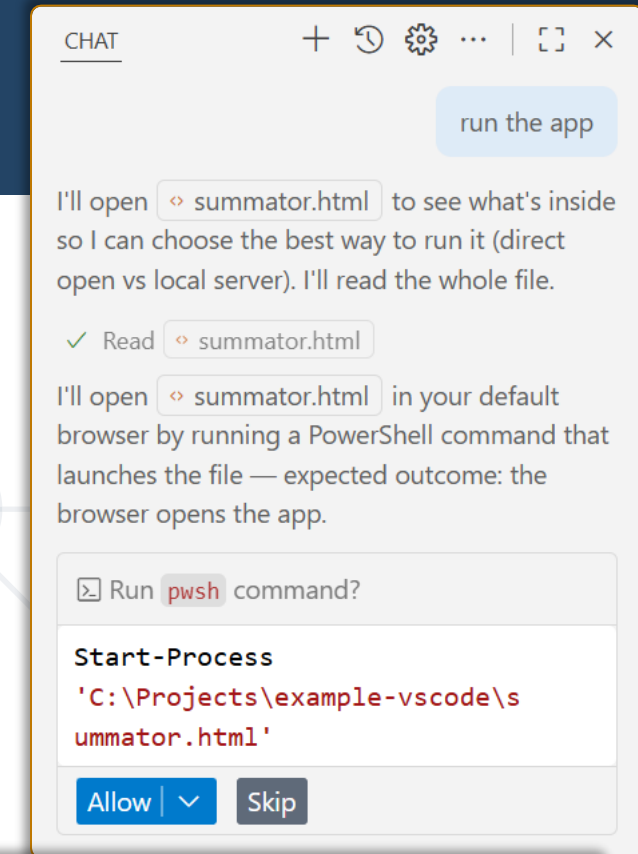
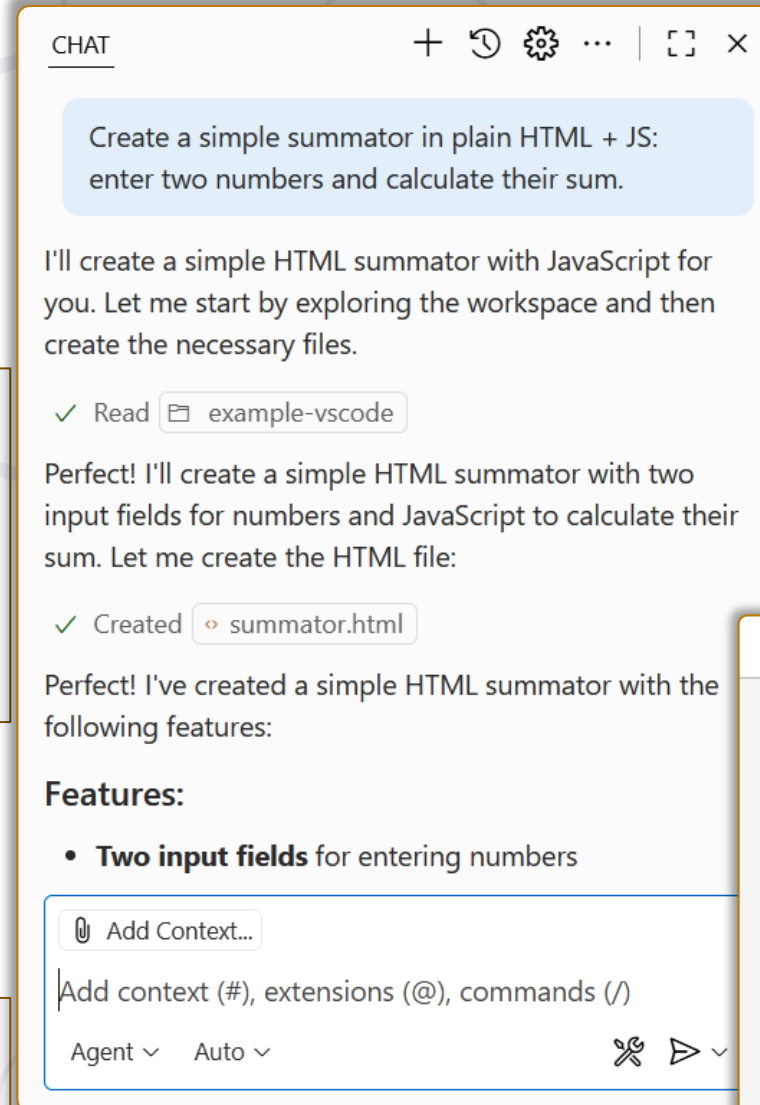
GitHub Copilot – Live Demo

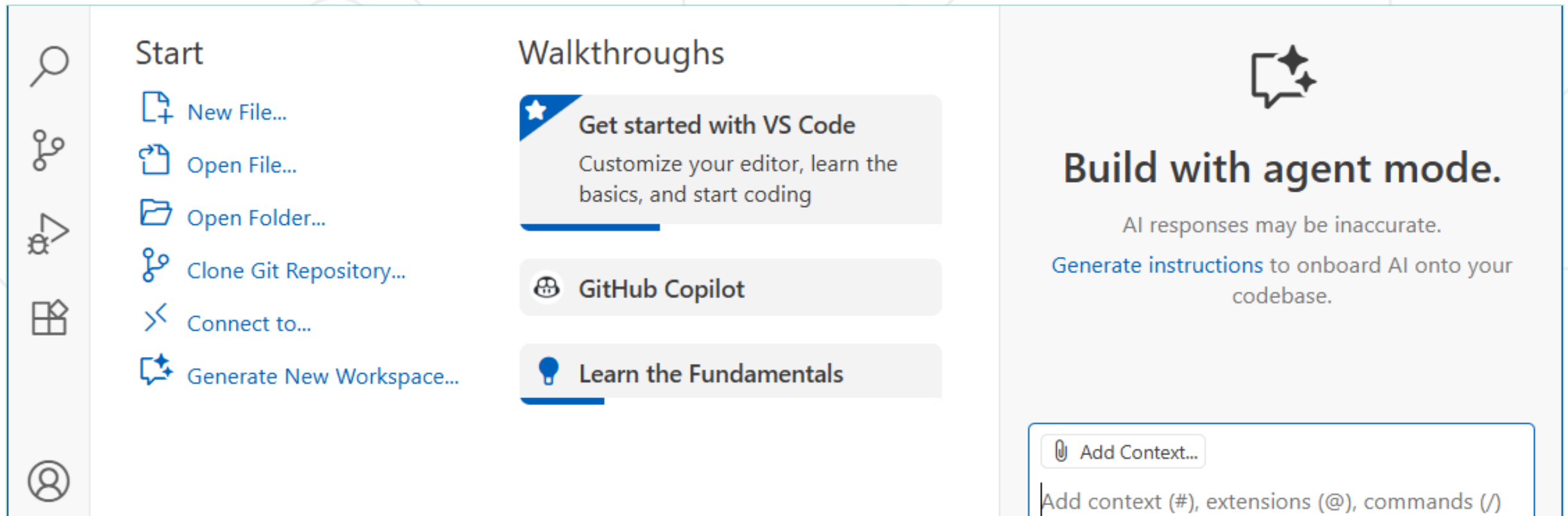
- Open VS Code
- Type this **prompt** in GitHub Copilot:

Create a **simple summator** in plain **HTML + JS**: enter two numbers and calculate their sum.

- Wait for the app to be generated → **start** it

run the app



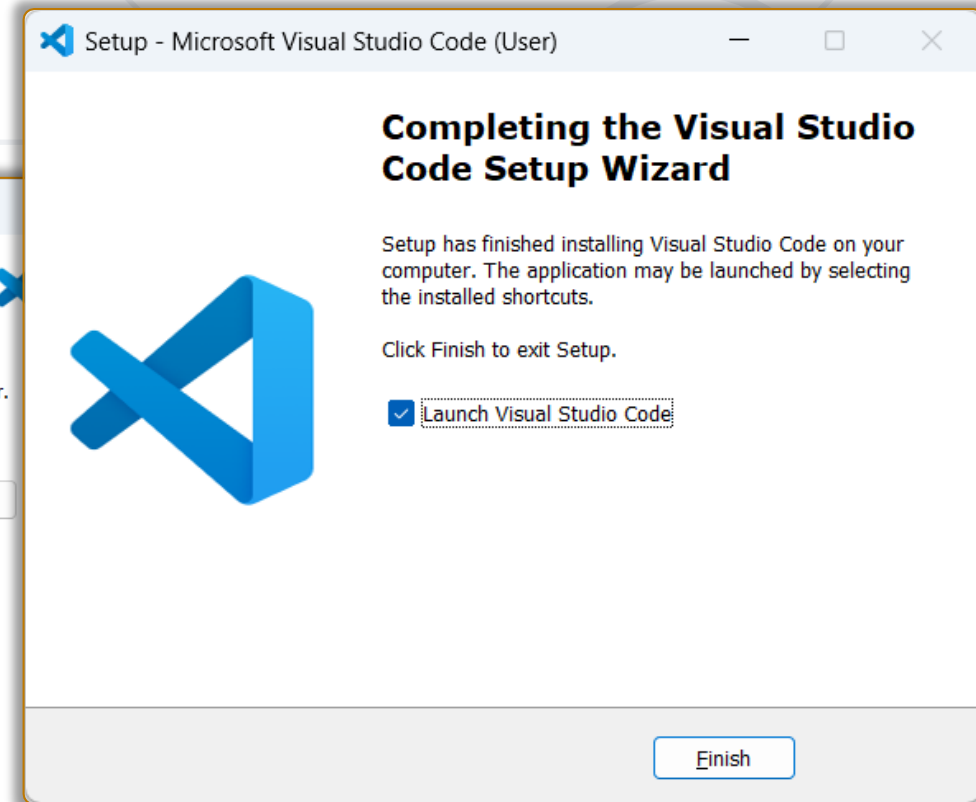
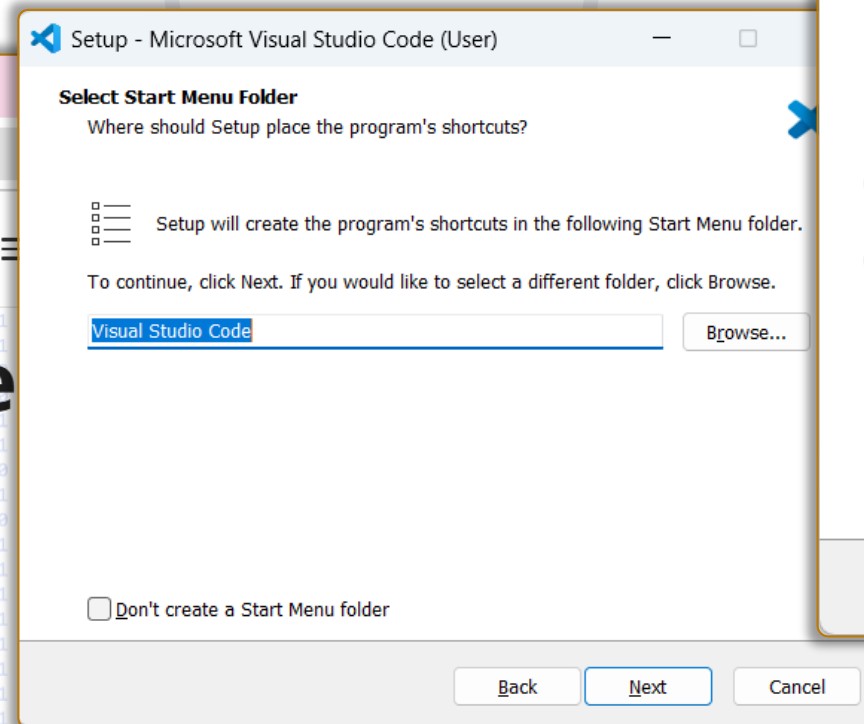
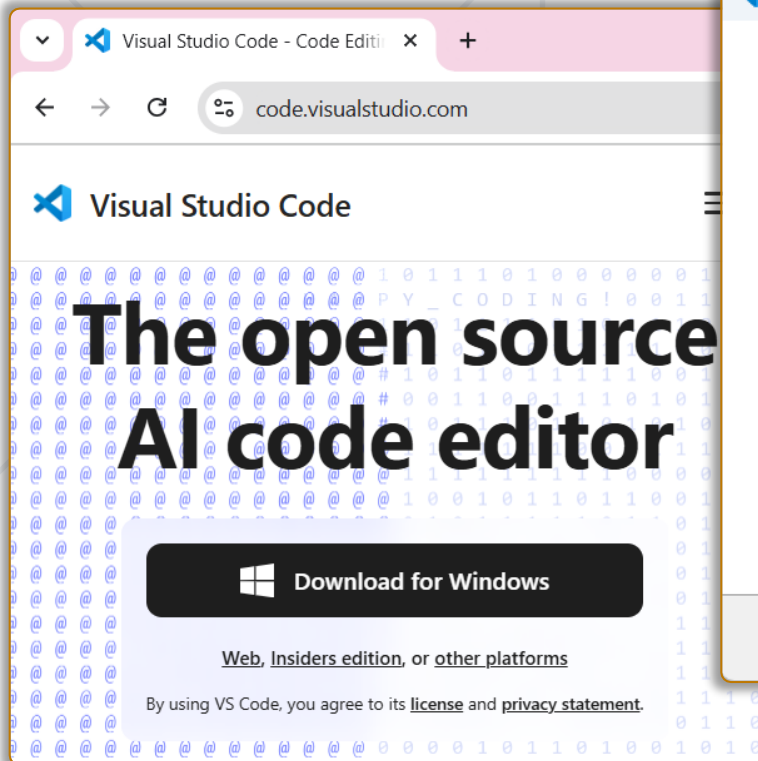


Installation, Alternatives, Pricing

VS Code + Extensions

Installing Visual Studio Code

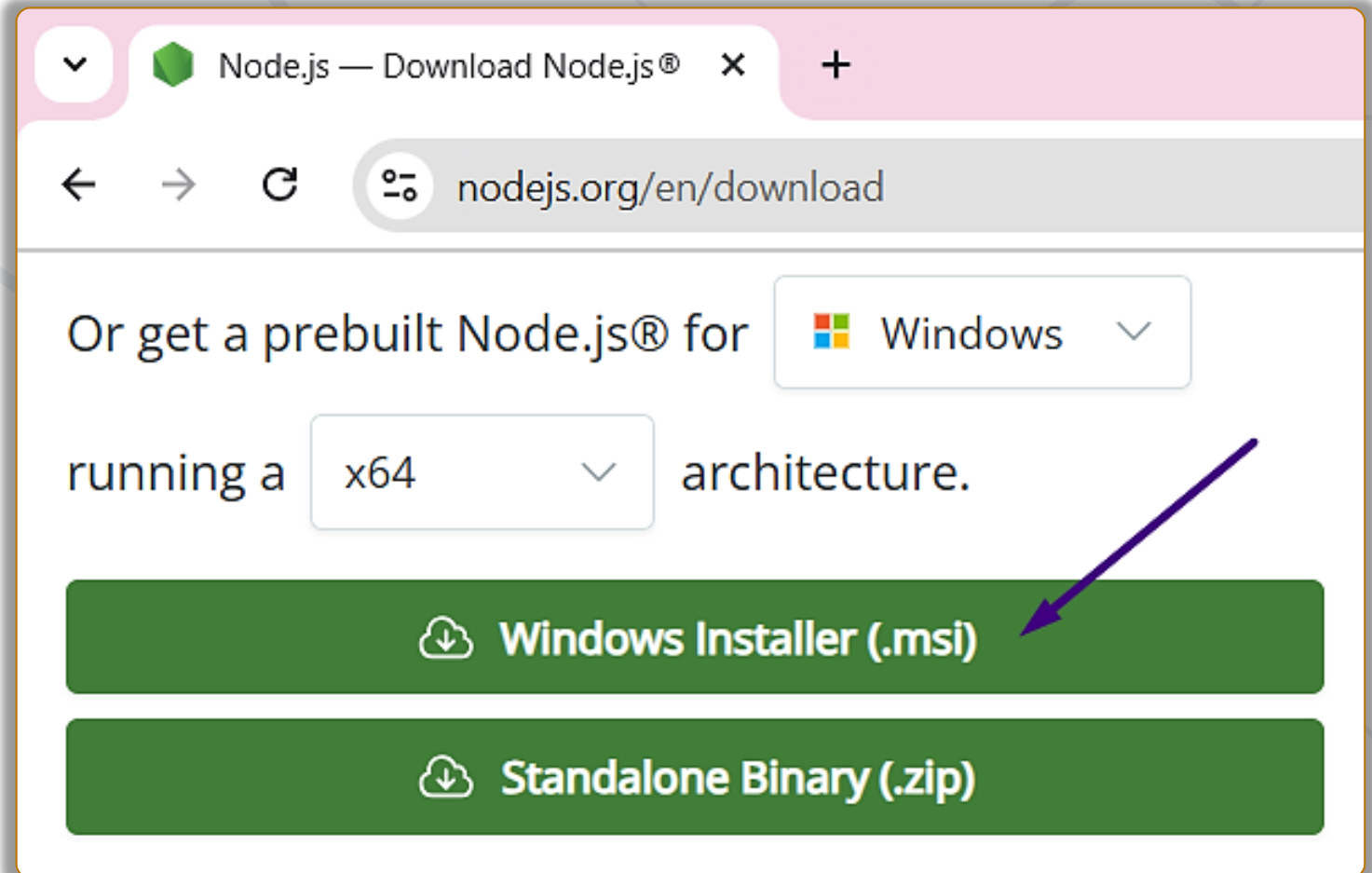
- **Download and install VS Code** (Visual Studio Code) from:
 - <https://code.visualstudio.com>



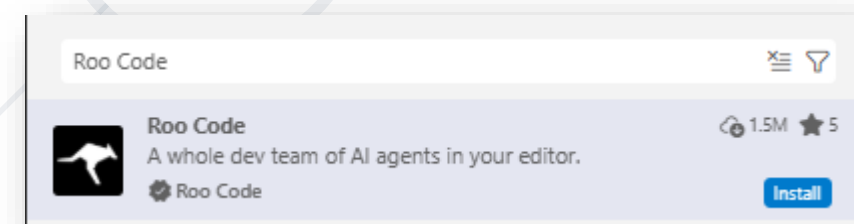
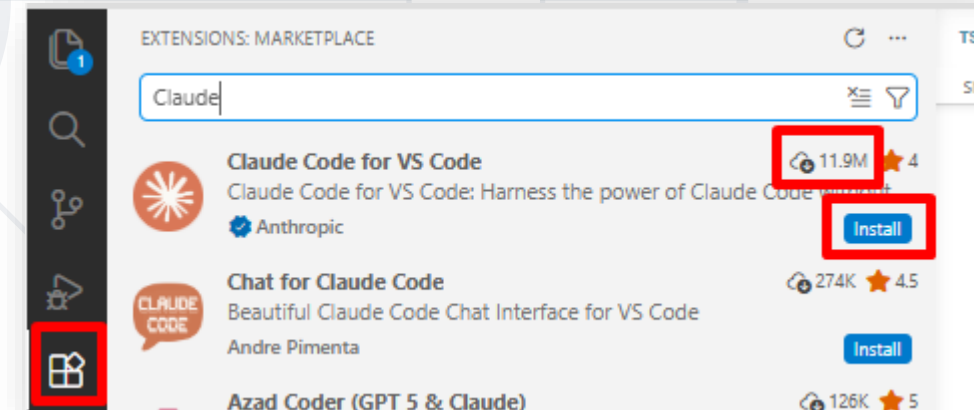
Install Node.js



- **Node.js** is a local JavaScript runtime for developers
- <https://nodejs.org>



- All **extensions** provide 90%+ **similar** functionality
- You can pick **any** option or **use multiple** alternatives at the same time
 - GitHub Copilot
 - Claude
 - Codex
 - Roo Code
 - Augment Code
 - Other alternatives



Licensing & Pricing

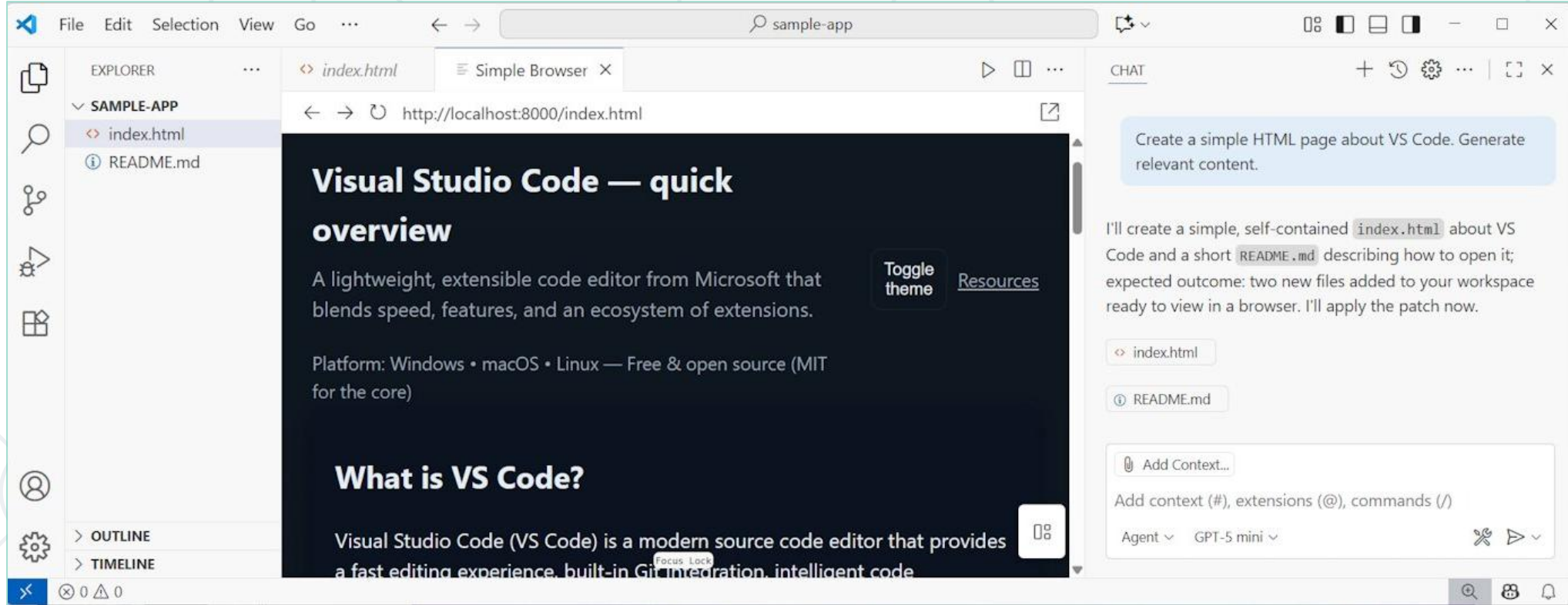
- **Paid** accounts with Free Tier
- **Free** extension – **paid models** (via OpenRouter or provider)
- Free extension with limited free models
- Advice: Start with free, connect via **OpenRouter**
- Constant changes
 - [LINK 1](#)
 - [LINK 2](#)

Free	Pro Most popular	Pro+
A fast way to get started with GitHub Copilot.	Accelerate workflows with GitHub Copilot.	Scale with agents and more models.
\$0 ^{USD}	\$10 ^{USD} per user / month	\$39 ^{USD} per user / month
Get started	Temporarily unavailable - read more	Temporarily unavailable - read more

The Unified Interface For LLMs

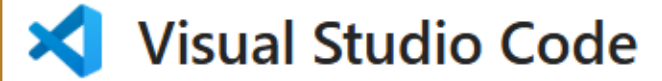
Better **prices**, better **uptime**, no subscriptions.

[Get API Key](#)[Explore Models](#) 



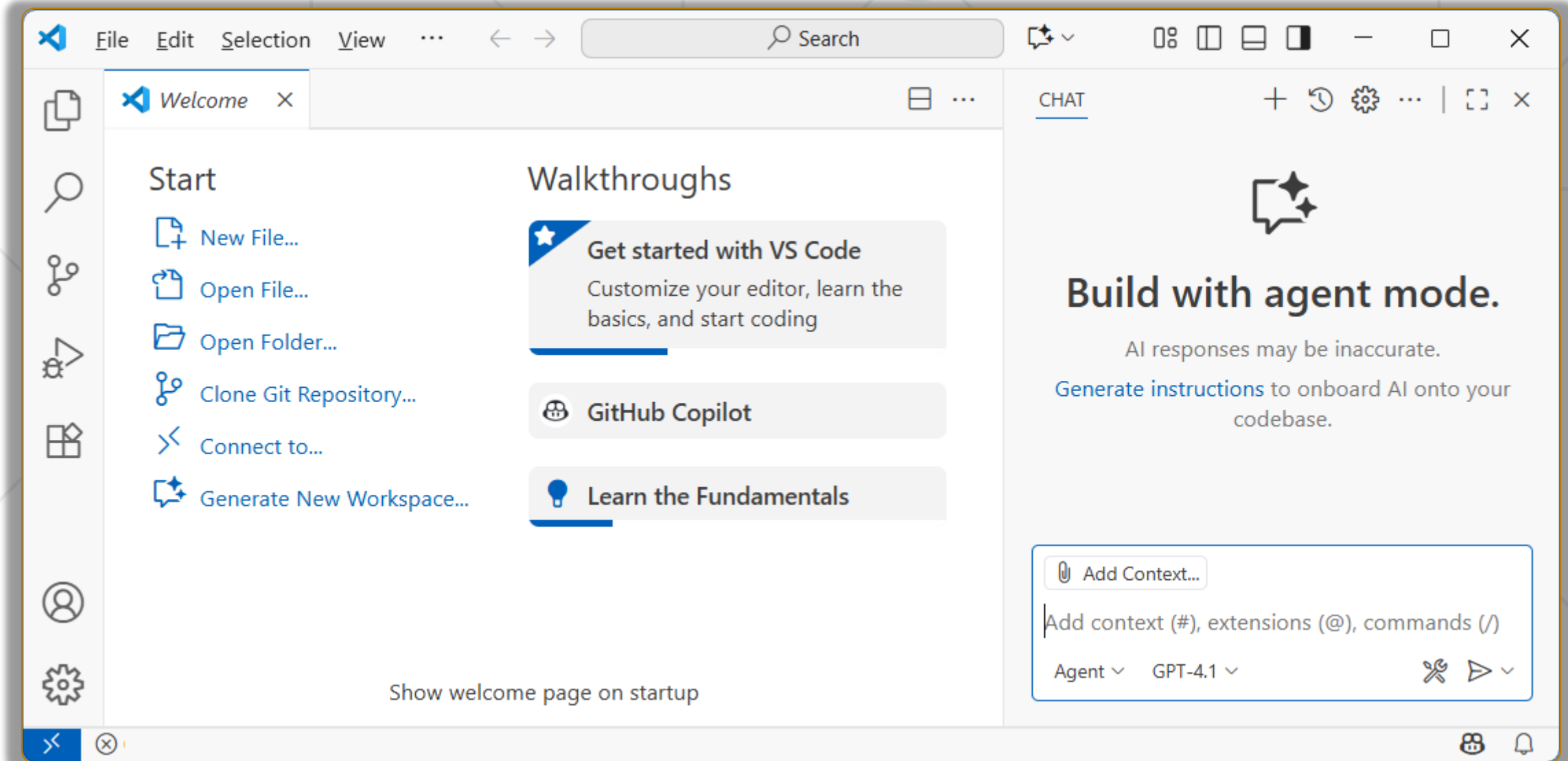
Your First App with AI-Assisted Coding

- **VS Code** is modern development IDE

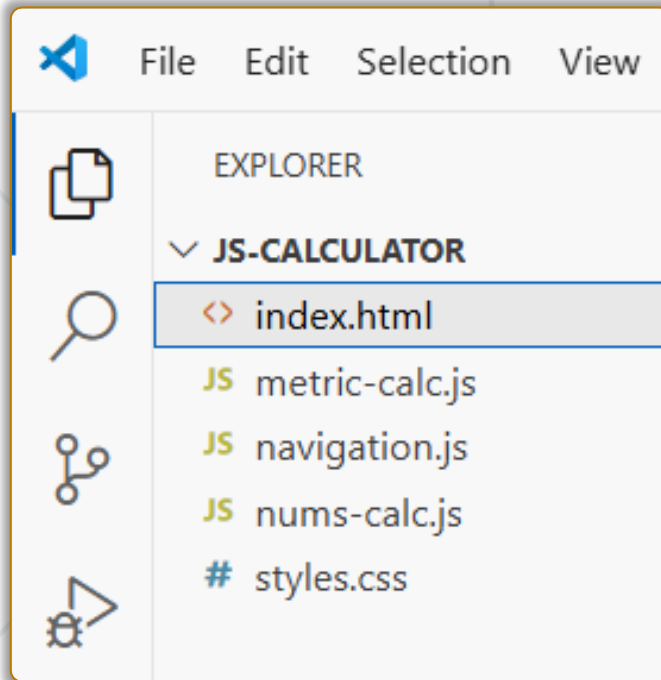


- In VS Code, developers **write** code, **execute** the code, **debug** and bug fix the code, **test** the code, bundle and **deploy** apps
- **Extensions** to support multiple languages / technologies
 - HTML, CSS, JavaScript, Python, C#, Java, PHP, C++, SQL, ...
 - Docker, Azure, AWS, Markdown, Jupiter, Supabase, ...
- Integrated **GitHub Copilot** for AI-assisted coding
- <https://code.visualstudio.com>

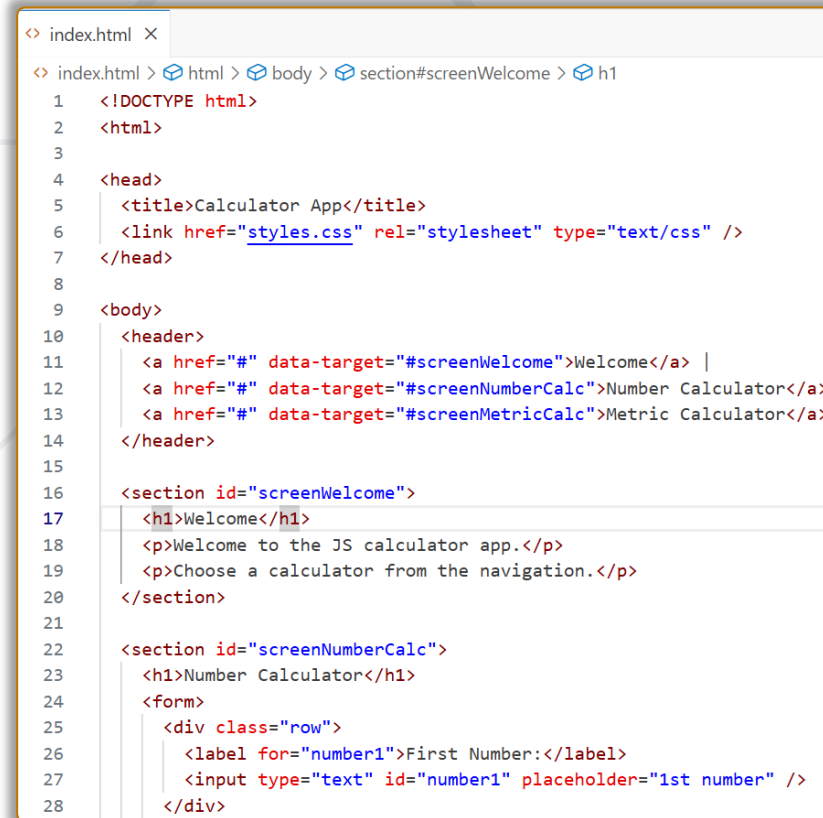
VS Code with GitHub Copilot



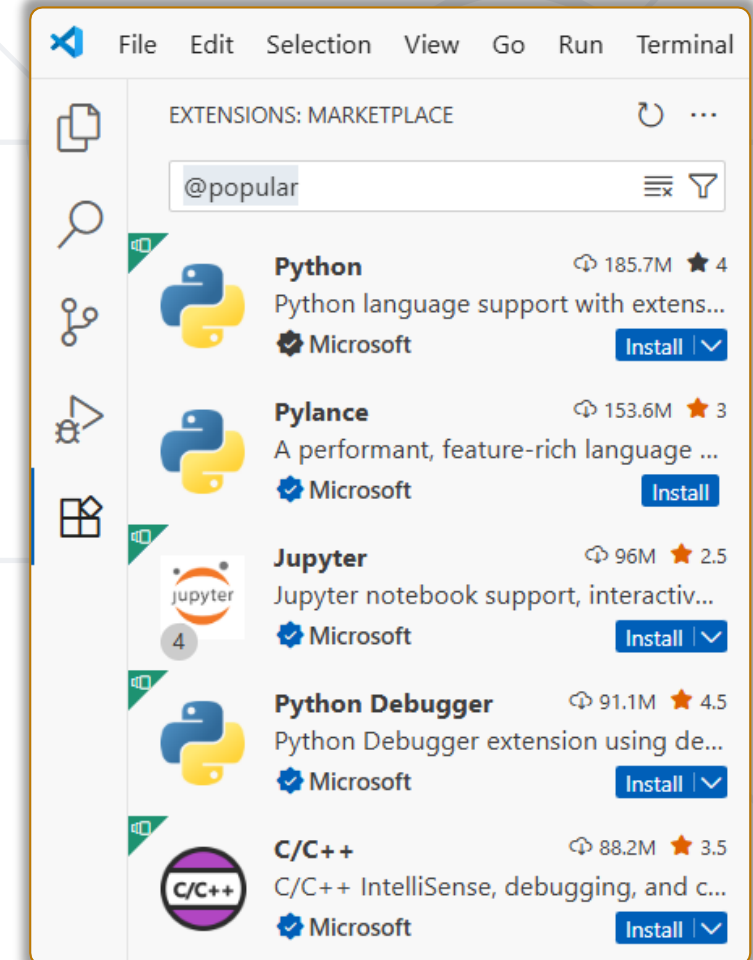
Code explorer



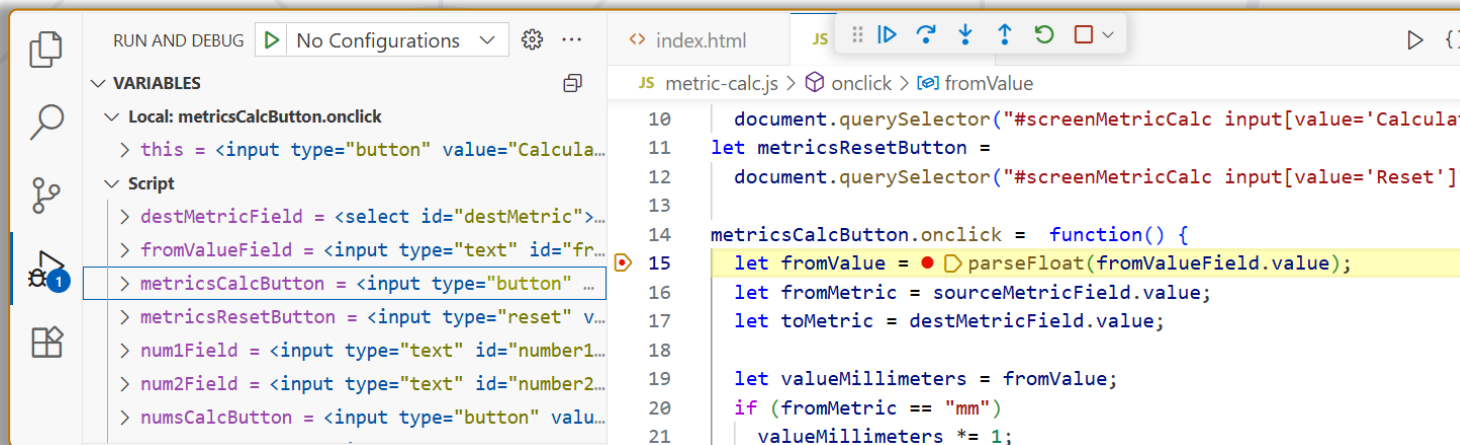
Code editor



Extensions

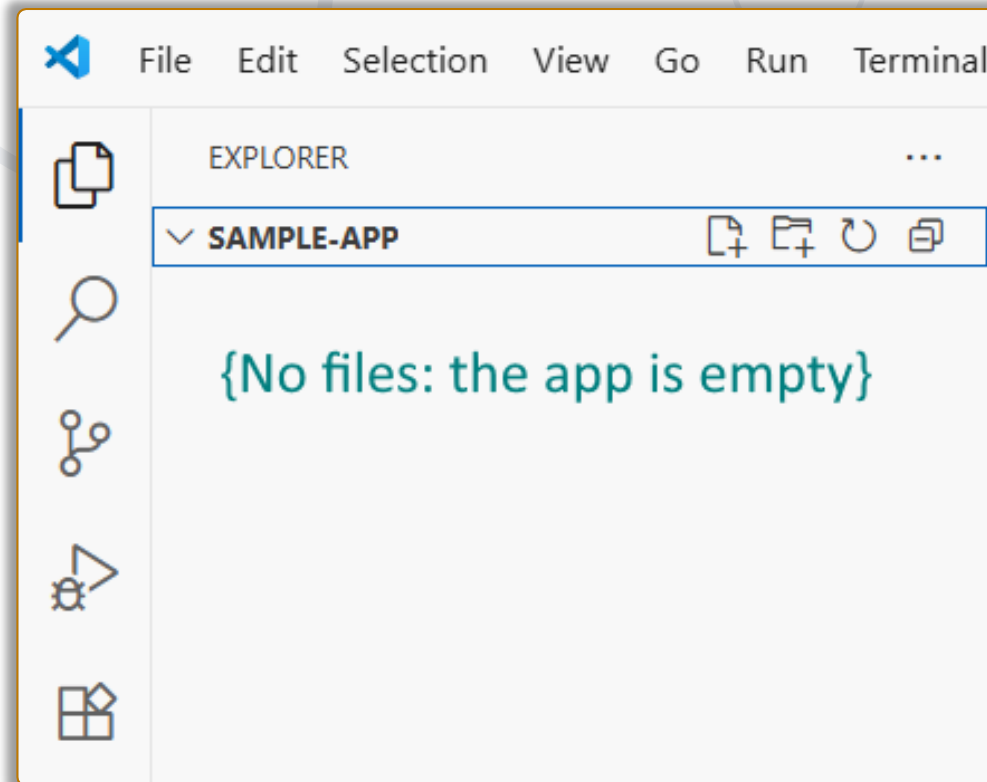
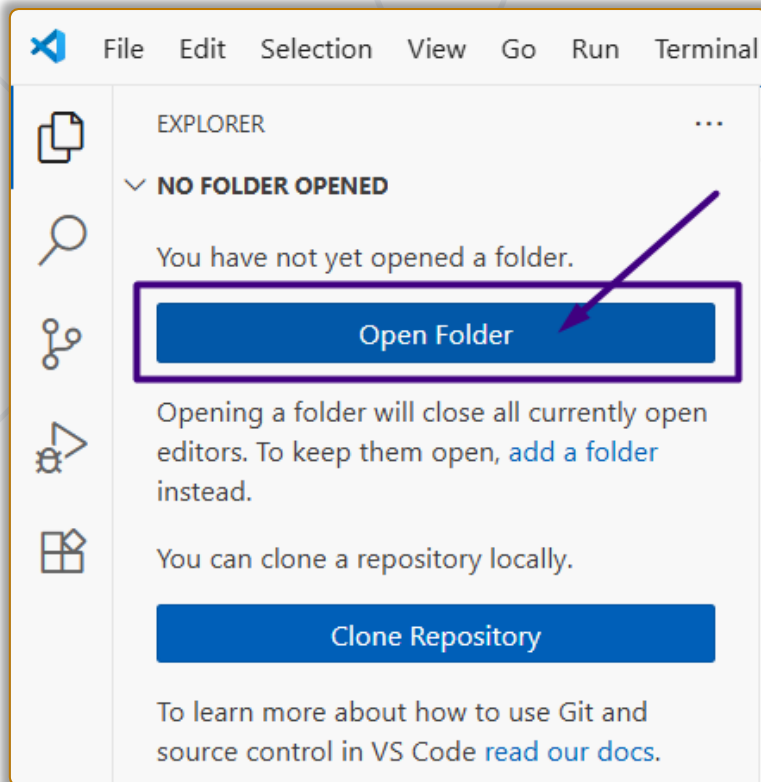


- **Debugger** for tracing code execution and finding bugs



Creating a New Project

- In VS Code **projects** are structured in **folders**
- Create a **new folder** and **open the folder** to start coding

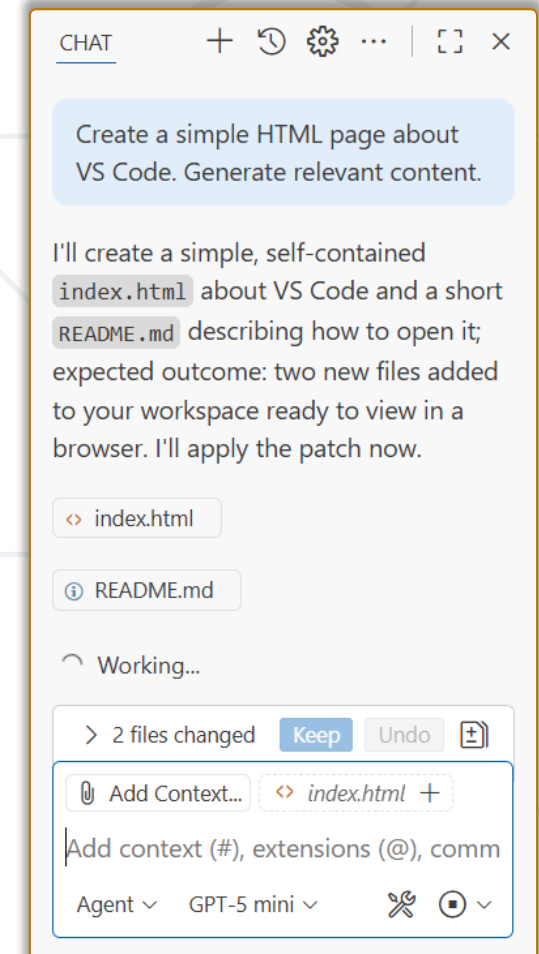
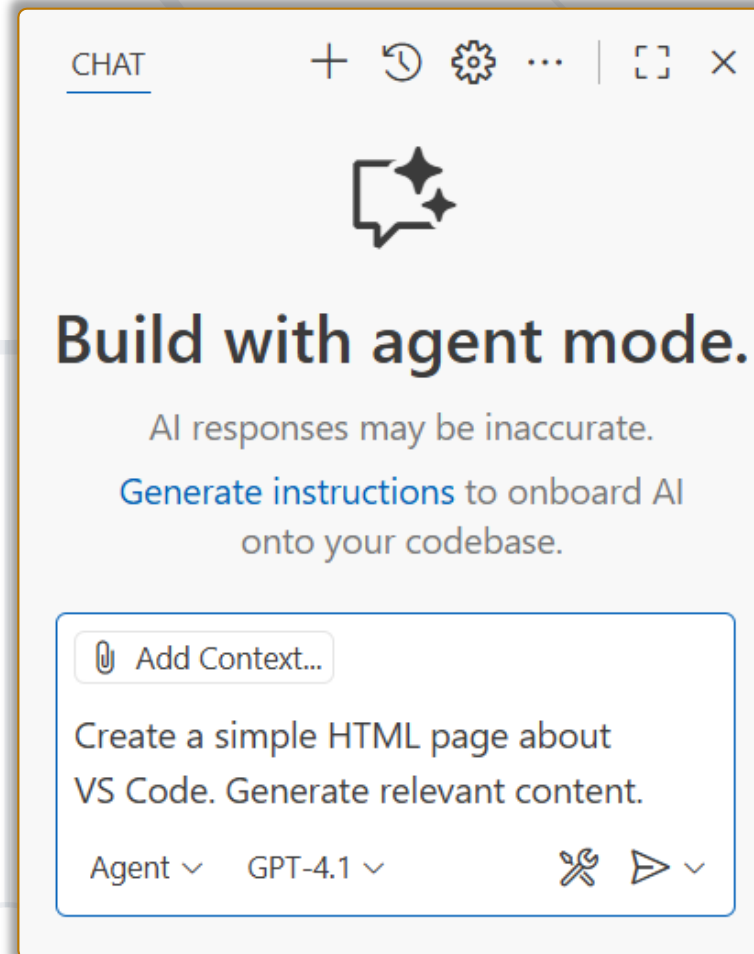


Creating a HTML Page

- Type this sample **prompt** in the GitHub Copilot chat:

Create a **simple HTML** page about VS Code.
Generate relevant content.

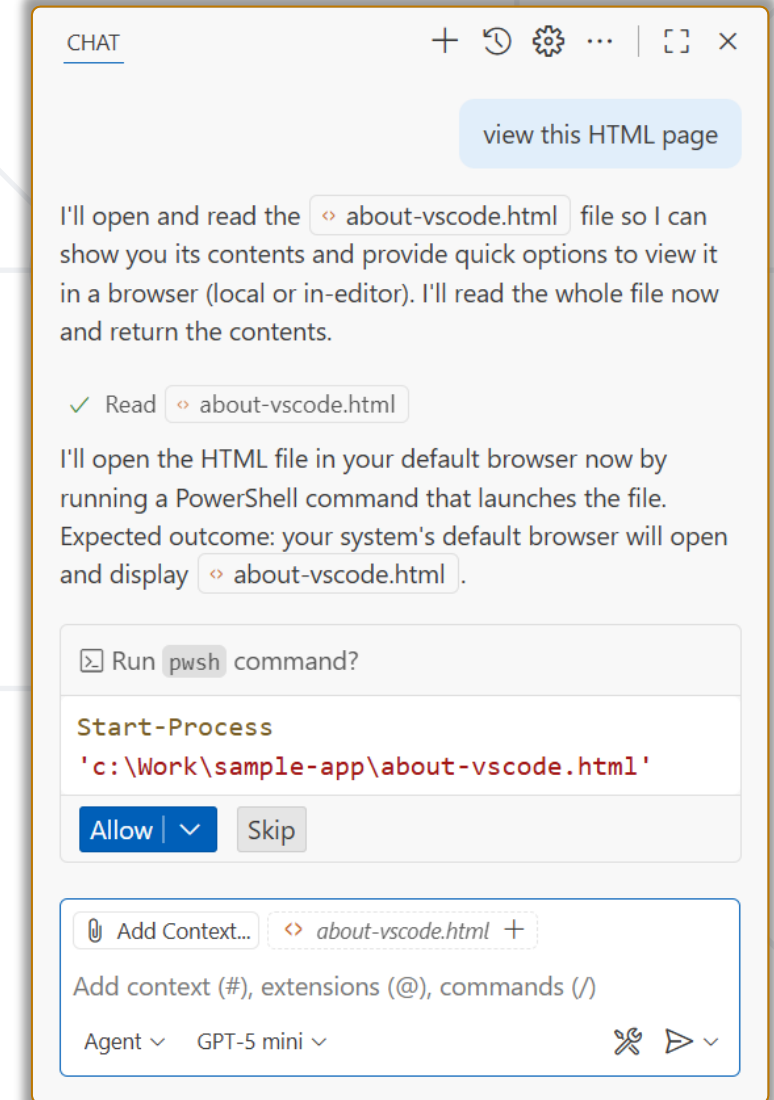
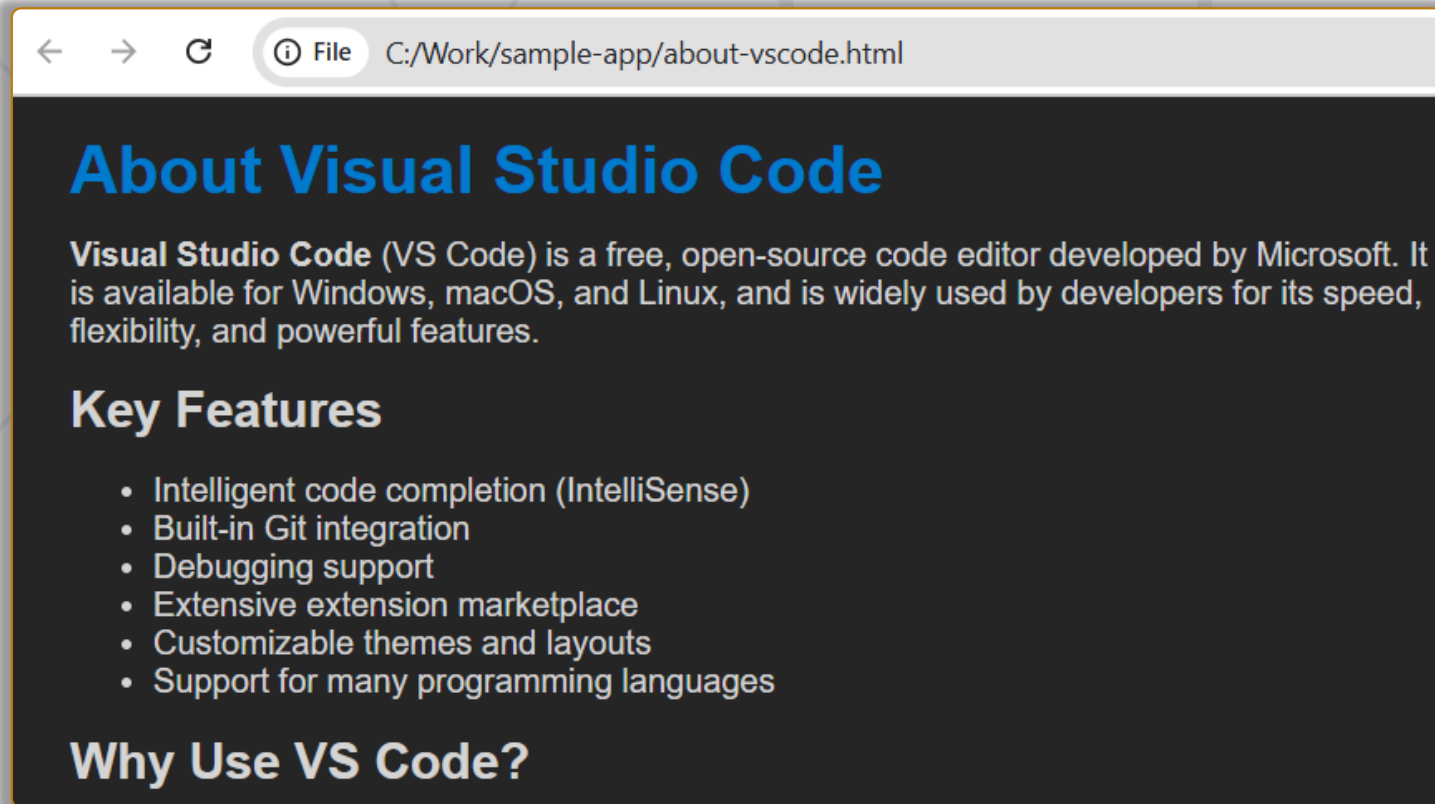
- Wait** for the Web page to be generated
- Accept** the proposed changes

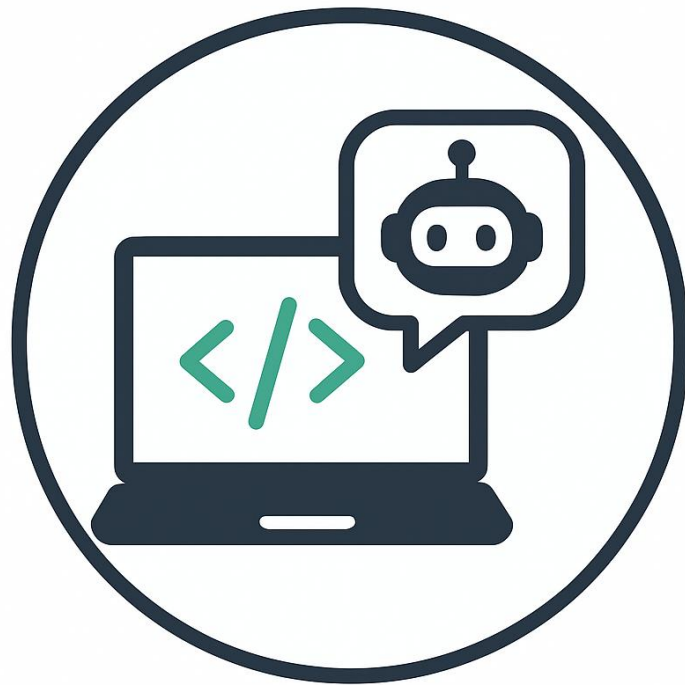


View the Generated HTML Page

- **Open** the page in the Web browser:

View this HTML page



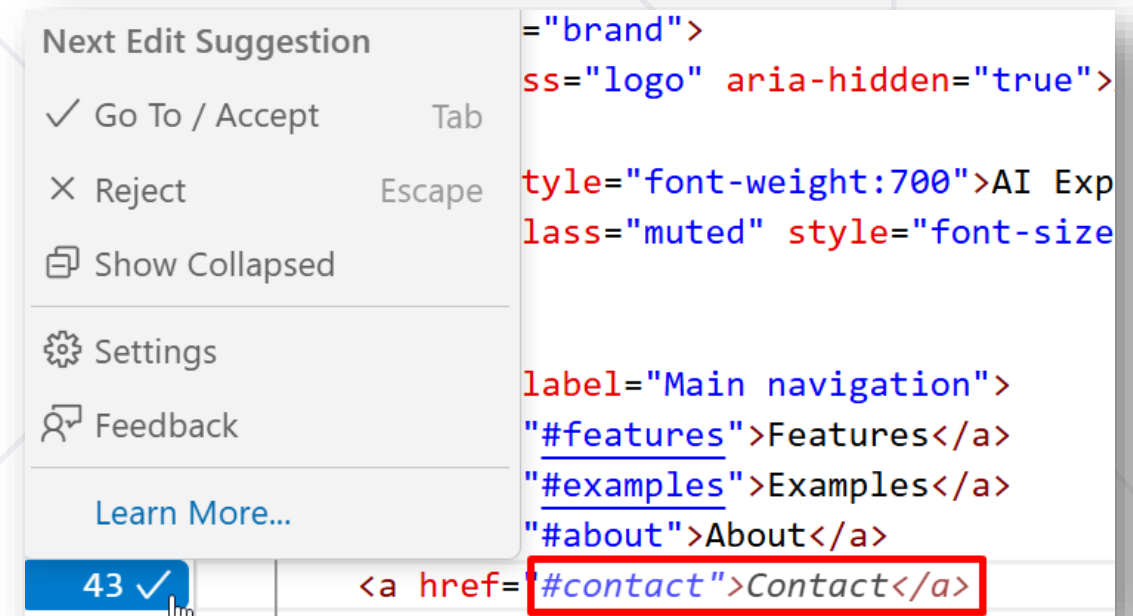


Inline Assistance

Your AI Pair Programmer Inside the Editor

Autocomplete

- **Suggests** code as you type
- **Learns** from file **context** and **patterns**
- **Reduces** boilerplate and repetitive typing
- Works across functions, classes, and full blocks
- **Improves speed** without breaking your flow



- Write a **comment** → Copilot turns it into **code**
- Supports **natural language** instructions
- Great for stubbing new functions quickly
- Can **generate** unit tests based on **requirements**
- **Saves time** on **routine** logic and structure

```
# Print top 5 most common products  
print("\nTop 5 most common products: ")
```

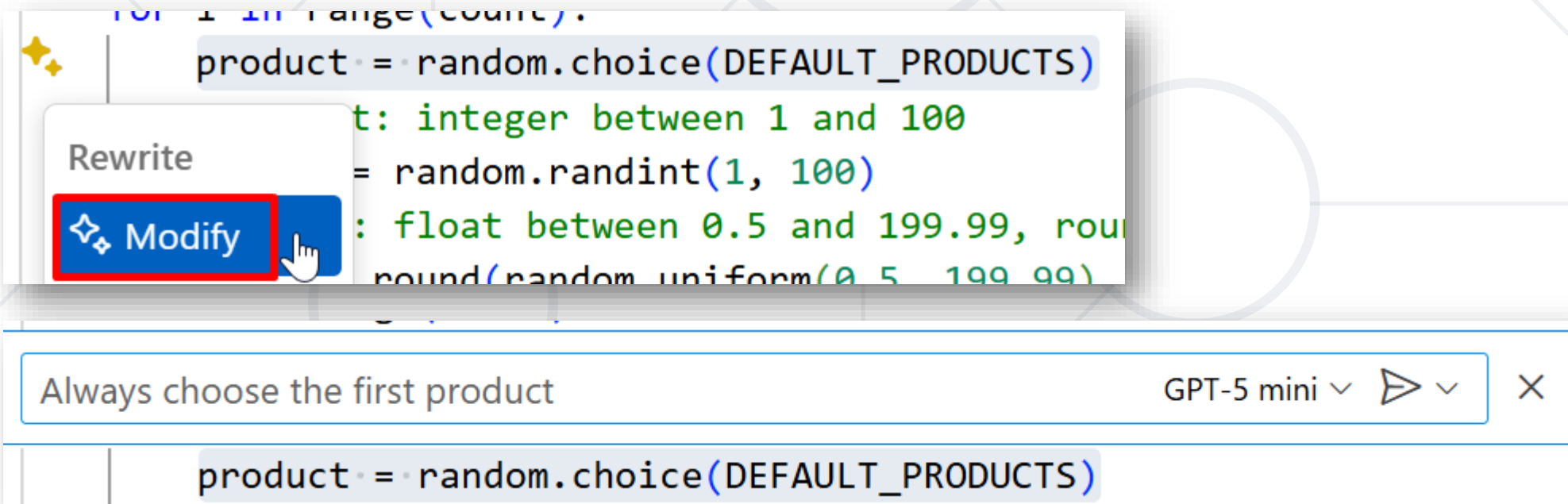
```
# Print top 5 most common products  
print("\nTop 5 most common products: ")  
print(df['product'].value_counts().head(5).to_string())
```

- **Summarizes** selected **code** in plain language
- Helps you **understand unfamiliar logic**
- Useful for **onboarding, debugging, or refactoring**

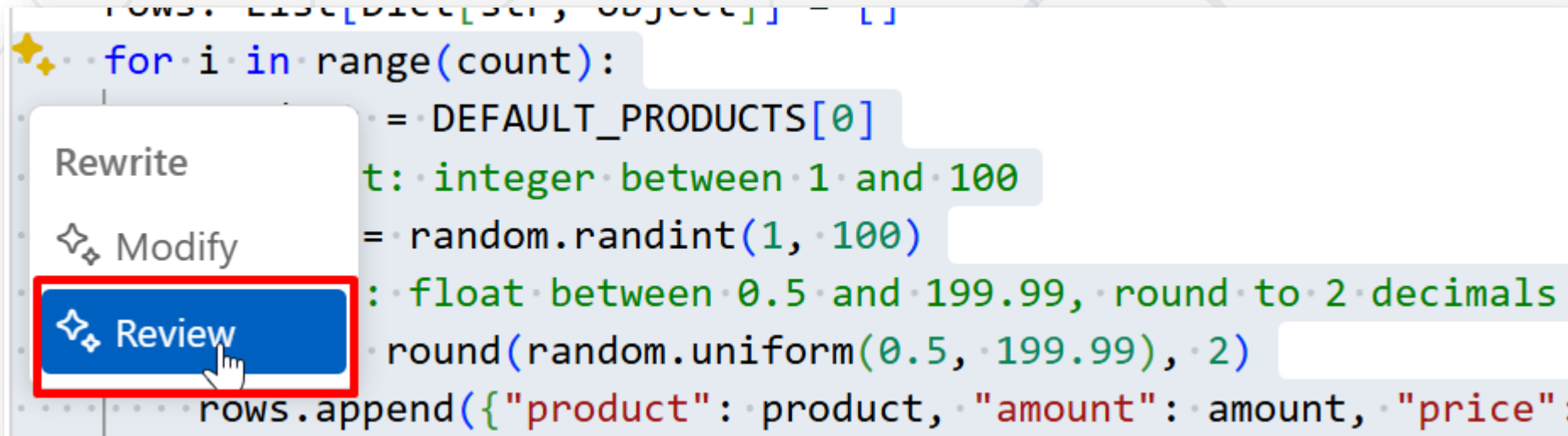
```
const a = document.body.animate(  
  [{ background: from }, { background: to }],  
  { duration: 420, easing: 'ease-in-out', fill: 'for'  
});  
a.onfinish = () => {  
  if (to === lightBg) {  
    root.style.setProperty('--bg', lightBg);  
    root.style.setProperty('--text', '■ #0b1220');  
  } else {  
    root.style.setProperty('--bg', darkBg);  
    root.style.setProperty('--text', '□ #e6eef8');
```

Find All References	Shift+Alt+F12
Find All Implementations	
Show Call Hierarchy	Shift+Alt+H
Add Selection to Chat	
Add File to Chat	
Open Inline Chat	Ctrl+I
Explain	

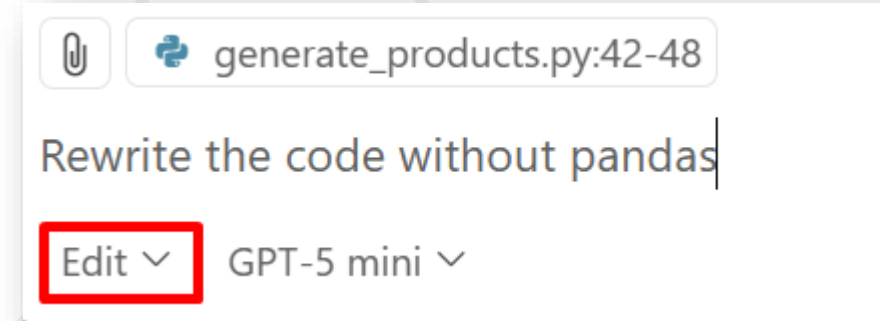
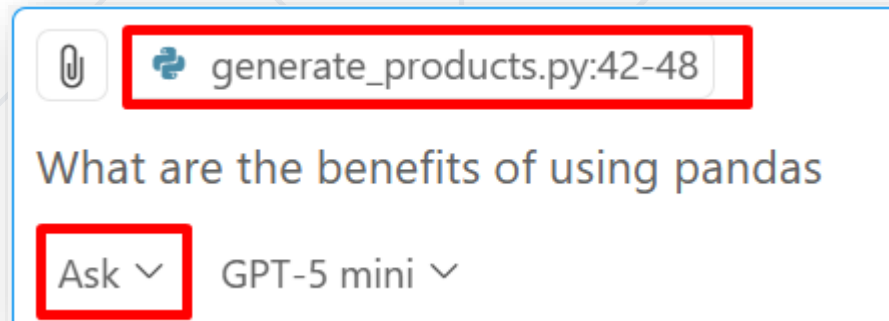
- **Select** code → tell Copilot how to **change** it
- Supports **optimization**, **cleanup**, and **rewrites**
- Can adjust **language**, **style**, or **complexity**



- **Flags** potential **issues** in selected code
- Suggests **improvements** and **best practices**
- Helps **catch bugs** early
- Can focus on **performance**, **readability**, or **security**



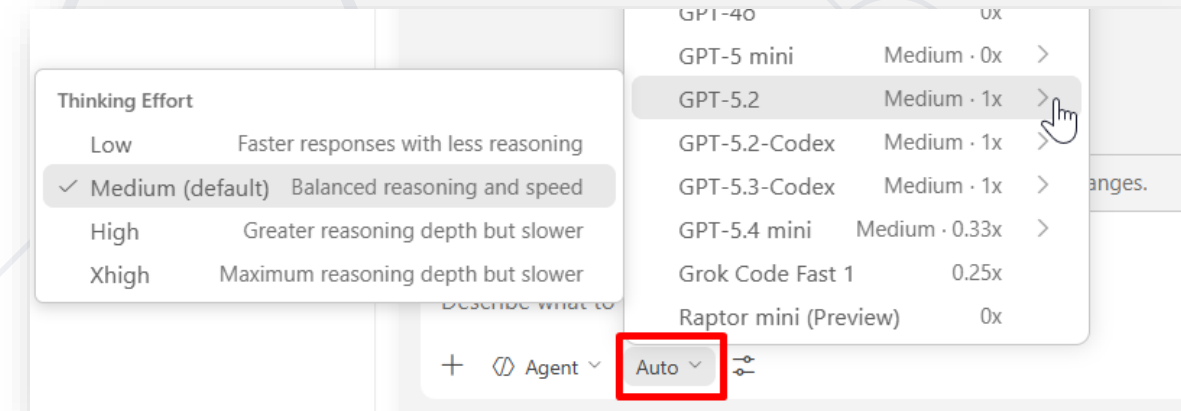
- Ask **questions** about your **code** or **project**
- Get **explanations**, **examples**, or **alternatives**
- Works with **selected code** or **entire files**



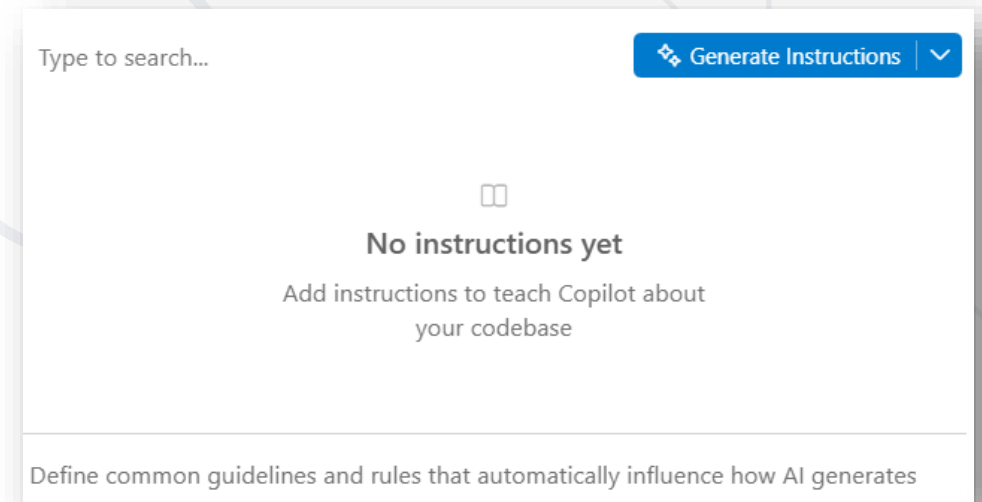
- Better **context** → better AI responses
- Use **#** to add context
- Add **files, folders, symbols**
- VS Code **auto-adds relevant** context
- **Drag & drop** or use **Add Context**



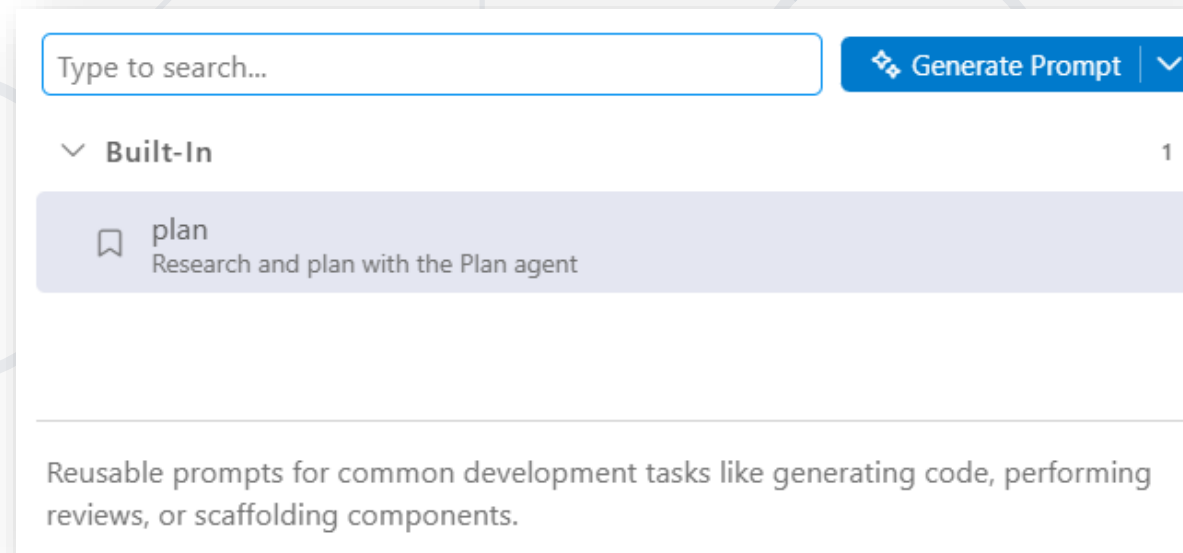
- **Provider Choice:** Toggle between models from OpenAI, Anthropic, and Google Gemini within the chat.
- **Auto-Selection:** Let Copilot dynamically route your request to the best model based on task complexity.
- **Reasoning Models:** Select "Thinking" models for complex architecture and deep debugging.
- **Speed vs. Power:** Choose lightweight models for fast boilerplate and high-performance models for logic-heavy refactoring.



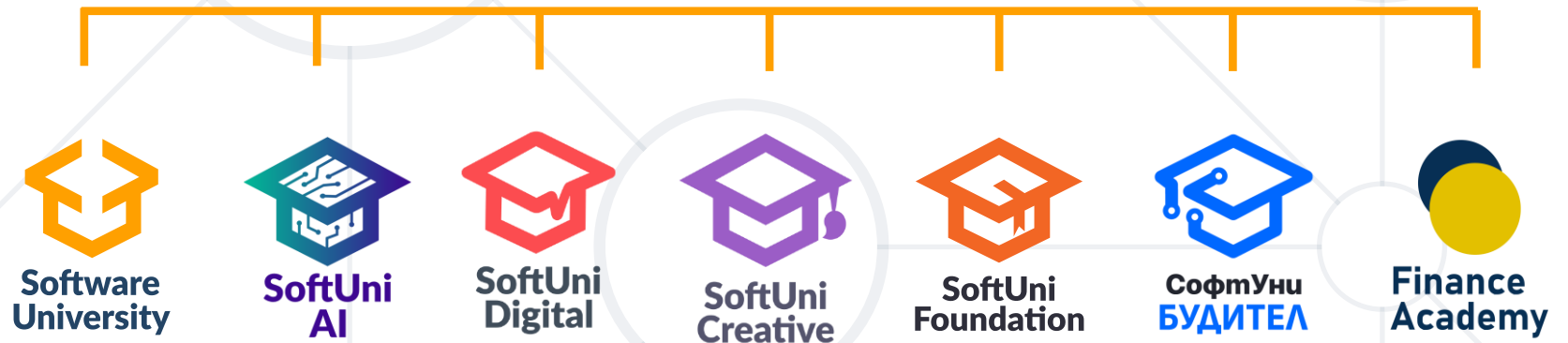
- **Configure** how Copilot behaves
- Set **custom rules** and **preferences**
- Keep style **consistent** across the project
- Acts as a **persistent** system **prompt**
- Helps maintain coding standards



- Save **reusable prompts** directly inside your repo
- Share instructions with the **whole team**
- Useful for **templates** and **recurring tasks**
- Copilot **automatically loads** them in Chat



Questions?



SoftUni Diamond Partners



- **Faster** coding with smart **suggestions** and **chat**
- Instantly **generate**, **explain**, and **refactor** code
- **Improve** code **quality** and **consistency** across the project



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